

Signed transfer from the absolute Jacobian formula

MathlibAnnex.BilipschitzOrientation.signed_area_transfer

theorem

Transfers an integrable function without first assuming that the Jacobian sign is constant.

Statement

Let $n \in \mathbb{N}_0$ and let $E = \mathbb{R}^n$ have the sup norm $\|\cdot\|_\infty$. Let $S, T \subseteq E$ be open, and let $f, g : E \rightarrow E$ be globally Lipschitz maps whose restrictions $f|_S : S \rightarrow T$ and $g|_T : T \rightarrow S$ are mutually inverse. Fix a number $\ell > 0$ such that

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \quad (x, x' \in E).$$

Write $Df(x) : E \rightarrow E$ for the Fréchet derivative of f at x when it exists, and set $Df(x) = 0$ at other points. Define the signed Jacobian by $J_f(x) = \det Df(x)$.

Define the source sign $s_f : E \rightarrow \{-1, 1\}$ and its transport $\sigma_f : E \rightarrow \{-1, 1\}$ by

$$s_f(x) = \begin{cases} 1, & J_f(x) \geq 0, \\ -1, & J_f(x) < 0, \end{cases} \quad \sigma_f(y) = s_f(g(y)).$$

In particular, a zero determinant gives sign $+1$, and $|\sigma_f(y)| = 1$ for every y .

All integrals use Lebesgue measure λ on these fixed coordinates. Let $\phi : E \rightarrow \mathbb{R}$ be integrable on T . Then

$$\int_S \phi(f(x)) J_f(x) d\lambda(x) = \int_T \phi(y) \sigma_f(y) d\lambda(y).$$

Assumptions

The integrability hypothesis is $\phi \in L^1(T, \lambda)$. No connectedness assumption is made on S or T .

The sets $S, T \subseteq E$ are open. The inverse and image conditions are

$$\begin{aligned} f(S) &\subseteq T, & g(T) &\subseteq S, \\ g(f(x)) &= x \quad (x \in S), & f(g(y)) &= y \quad (y \in T). \end{aligned}$$

The global bounds use specified constants $C_f, C_g \geq 0$ and $\ell > 0$:

$$\ell \|x - x'\|_\infty \leq \|f(x) - f(x')\|_\infty \leq C_f \|x - x'\|_\infty \quad (x, x' \in E),$$

$$\|g(y) - g(y')\|_\infty \leq C_g \|y - y'\|_\infty \quad (y, y' \in E).$$

The estimates hold on all of E , not just on S or T . The inverse identities are required only on S and T .

Conclusion

The factor on the target is the sign of the source Jacobian evaluated at the inverse point $g(y)$. It may vary with y . Both integrands are integrable on their respective sets.

The total derivative is measurable, its determinant is measurable, and the displayed two-valued sign is measurable. Since g is continuous, $\sigma_f = s_f \circ g$ is measurable as well. These facts justify the integrability assertion for $\phi\sigma_f$ used in the proof. The two global Lipschitz constants are exactly those supplied for f and g in the assumptions.

Proof route

Discard a null subset of S and a null subset of T , apply the injective absolute area formula, and use $g(f(x)) = x$ to restore the signed determinant.

▼ Proof steps

1. **Identify the two null complements.** Define

$$N = \{x \in E : f \text{ is not differentiable at } x\}, \quad G = S \setminus N.$$

Then

$$\lambda(N) = 0, \quad \lambda(S \setminus G) = 0, \quad \lambda(f(N)) = 0.$$

The first equality is Rademacher's theorem for the globally Lipschitz map f on the finite-dimensional space E . The second follows from $S \setminus G \subseteq N$. The third is preservation of null sets by an equal-dimensional Lipschitz map, with this same f and N ; it is also recorded by the target-exception lemma. The set G is measurable because S and the differentiability set are measurable.

The inverse identities give

$$f(G) \subseteq T, \quad T \setminus f(G) \subseteq f(N), \quad \lambda(T \setminus f(G)) = 0.$$

Indeed, if $y \in T \setminus f(G)$, then $g(y) \in S$ and $f(g(y)) = y$. Hence $g(y) \notin G$, so $g(y) \in N$ and $y \in f(N)$.

2. **Apply the absolute formula to the signed function.** Put $\psi(y) = \phi(y)\sigma_f(y)$. The sign σ_f is measurable and $|\psi| = |\phi|$, so ψ is integrable on T and on $f(G)$. The formula to be used is

$$\int_{f(G)} \psi(y) d\lambda(y) = \int_G |J_f(x)| \psi(f(x)) d\lambda(x).$$

For the injective absolute Jacobian formula, the domain is the measurable set G , the map is f , the derivative at each $x \in G$ is $Df(x)$ (also a derivative within G), and the measure is λ . Injectivity on G follows from $g(f(x)) = x$ on S . With these same inputs, the corresponding integrability equivalence gives integrability of the right-hand integrand.

3. **Substitute the inverse and remove the null complements.** For $x \in G$,

$$\sigma_f(f(x)) = s_f(g(f(x))) = s_f(x), \quad s_f(x)|J_f(x)| = J_f(x).$$

The second identity is the signed determinant identity at this differentiability point. Therefore

$$|J_f(x)|\psi(f(x)) = |J_f(x)|\phi(f(x))\sigma_f(f(x)) = \phi(f(x))J_f(x).$$

Substituting this equality into Step 2 and using the null complements from Step 1 gives

$$\begin{aligned} \int_S \phi(f(x))J_f(x) d\lambda(x) &= \int_G \phi(f(x))J_f(x) d\lambda(x) \\ &= \int_{f(G)} \phi(y)\sigma_f(y) d\lambda(y) \\ &= \int_T \phi(y)\sigma_f(y) d\lambda(y). \end{aligned}$$

The first and last replacements are justified by $\lambda(S \setminus G) = 0$ and $\lambda(T \setminus f(G)) = 0$, respectively. The source integrability obtained in Step 2 is preserved by the first replacement.

Main citations

- Exact declaration and its source — `MathlibAnnex.BilipschitzOrientation.signed_area_transfer`
- Injective absolute area formula for an observable — `MeasureTheory.integral_image_eq_integral_abs_det_fderiv_smul`
- Integrability under that same absolute formula — `MeasureTheory.integrableOn_image_iff_integrableOn_abs_det_fderiv_smul`
- Forward global Lipschitz witness — `MathlibAnnex.BilipschitzOrientation.BiLipschitzOpenData.lipschitz_f`
- Inverse global Lipschitz witness — `MathlibAnnex.BilipschitzOrientation.BiLipschitzOpenData.lipschitz_g`
- Lipschitz images preserve null sets in equal dimension — `MathlibAnnex.BilipschitzOrientation.volume_image_eq_zero_of_lipschitzWith`
- Total Jacobian sign, including the zero convention — `MathlibAnnex.BilipschitzOrientation.domainJacobianSign`
- Transport of the sign by the specified inverse — `MathlibAnnex.BilipschitzOrientation.targetJacobianSign`
- Recovering the signed determinant from its absolute value — `MathlibAnnex.BilipschitzOrientation.det_eq_domainSign_mul_abs`
- The image of the nondifferentiability set is null — `MathlibAnnex.BilipschitzOrientation.target_exceptional_null`
- Rademacher for globally Lipschitz maps in finite dimension — `LipschitzWith.ae_differentiableAt`

Lean source signature (exact)

```
theorem signed_area_transfer {n : ℕ} (D : BiLipschitzOpenData n)
  {ϕ : (Fin n → ℝ) → ℝ} (_hϕ : IntegrableOn ϕ D.target) :
  ∫ x in D.source, ϕ (D.f x) * (fderiv ℝ D.f x).det ∂volume =
  ∫ y in D.target, ϕ y * targetJacobianSign D y ∂volume
```

In the source	Mathematical meaning
<code>(D : BiLipschitzOpenData n)</code>	The record described in this Card: open $S, T \subseteq E = \mathbb{R}^n$, maps $f, g : E \rightarrow E$ restricting to inverse bijections $S \leftrightarrow T$, global upper Lipschitz bounds C_f, C_g , and global lower bound $\ell \ x - x'\ _\infty \leq \ f(x) - f(x')\ _\infty$ with $\ell > 0$. The source name <code>D</code> is not the derivative symbol.
<code>D.source; D.target; D.f; D.g</code>	The local mathematical objects S, T, f, g , respectively, throughout this Card.
<code>{ϕ : (Fin n → ℝ) → ℝ} (hϕ : IntegrableOn ϕ D.target)</code>	The scalar weight ϕ is integrable on T for Lebesgue measure. The underscore in the hypothesis name does not remove this assumption.
<code>(fderiv ℝ D.f x).det</code>	The signed Jacobian $J_f(x) = \det Df(x)$, with the total derivative convention stated in the body.
<code>targetJacobianSign D</code>	The scalar function $\sigma_f(y) = s_f(g(y))$, where $J_f(x) = \det Df(x)$ and $s_f(x) = 1$ for $J_f(x) \geq 0$, $s_f(x) = -1$ for $J_f(x) < 0$. Zero has sign $+1$ under the cited definitions.

In the source	Mathematical meaning
<code>∫ x in D.source, ϕ (D.f x) * (fderiv ℝ D.f x).det ∂volume</code>	The source integral $\int_S \phi(f(x)) J_f(x) d\lambda(x)$, with fixed coordinate Lebesgue measure λ .
<code>∫ y in D.target, ϕ y * targetJacobianSign D y ∂volume</code>	The equal target integral $\int_T \phi(y) \sigma_f(y) d\lambda(y)$. Both the weight and transported sign are evaluated at the same y .

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 23–29:

```
noncomputable section

open Set MeasureTheory Filter
open scoped ENNReal NNReal Topology

namespace MathlibAnnex
namespace BilipschitzOrientation
```

▼ Exact content identity

Declaration: `MathlibAnnex.BilipschitzOrientation.signed_area_transfer`

Accepted content SHA-256: 8abd331b0ff7b7ac74b238db75a3acf83fcbc42fa739d87353c82fce262d9802

Accepted source guide SHA-256: 3aca907950df5947776741b663d8284043b940fd54d211c7f3444e187ab27b5e

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73